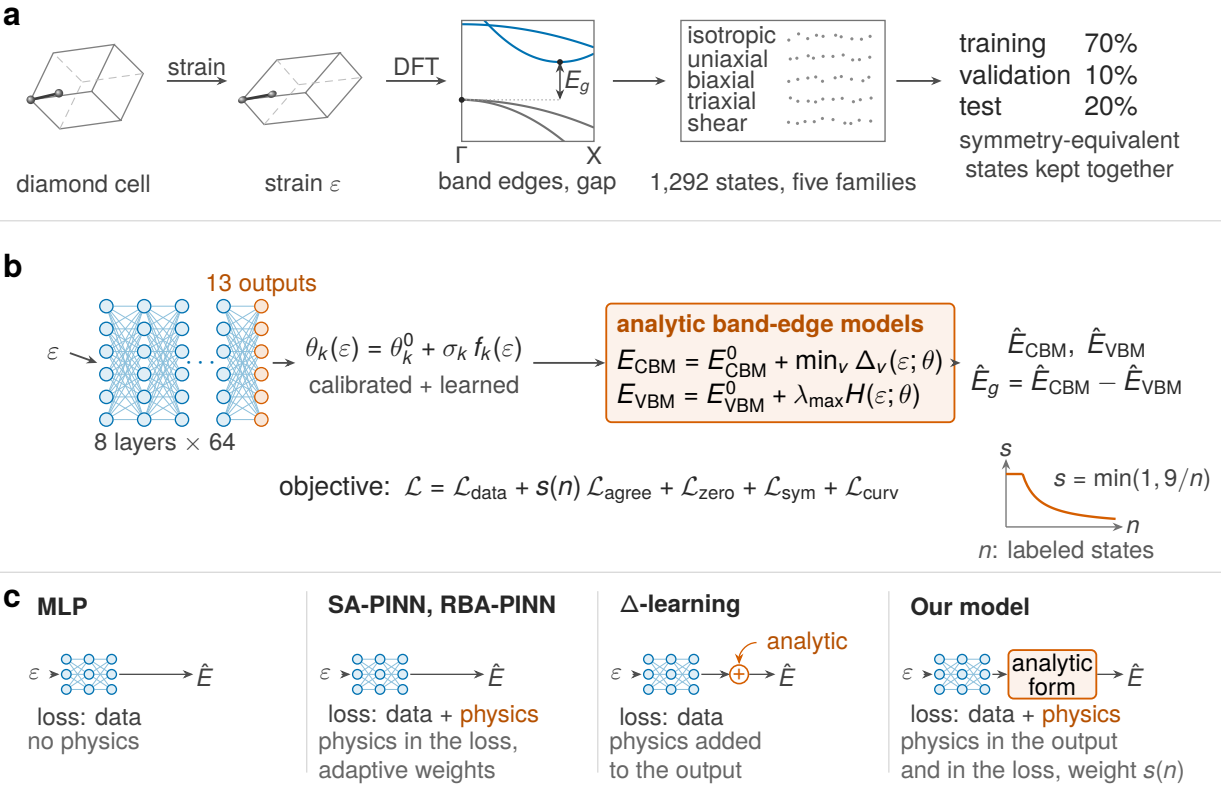

a
strain
DFT
E_g
Γ
X
isotropic
uniaxial
biaxial
triaxial
shear
training 70%
validation 10%
test 20%
symmetry-equivalent states kept together
diamond cell
strain ε
band edges, gap
1,292 states, five families
b
13 outputs
ε
8 layers × 64
$\theta_k(\varepsilon) = \theta_k^0 + \sigma_k f_k(\varepsilon)$
calibrated + learned
analytic band-edge models
$E_{CBM} = E^0_{CBM} + \min_v \Delta_v(\varepsilon;\theta)$
$E_{VBM} = E^0_{VBM} + \lambda_{max} H(\varepsilon;\theta)$
$\hat{E}_{CBM}$, $\hat{E}_{VBM}$
$\hat{E}_g = \hat{E}_{CBM} - \hat{E}_{VBM}$
objective: $\mathcal{L} = \mathcal{L}_{data} + s(n)\,\mathcal{L}_{agree} + \mathcal{L}_{zero} + \mathcal{L}_{sym} + \mathcal{L}_{curv}$
s
$s = \min(1, 9/n)$
n
n: labeled states
c
MLP
loss: data
no physics
SA-PINN, RBA-PINN
loss: data + physics
physics in the loss,
adaptive weights
Δ-learning
analytic
loss: data
physics added
to the output
Our model
analytic form
loss: data + physics
physics in the output
and in the loss, weight s(n)